

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from some distribution.

Consider testing

H_0 : The sample comes
from a particular,
specified distribution

versus

H_1 : “Not H_0 ”

Examples of H_0 :

- The sample comes from a binomial distribution with parameters 8 and 0.2.
- The sample comes from a $N(0, 1)$ distribution.

- The sample comes from this distribution:

x	0	1	2	3
$P(X = x)$	0.2	0.4	0.1	0.3

Here are some sampled values:

1, 1, 2, 0, 1, 1, 1, 1, 3, 3,
1, 2, 0, 3, 1, 1, 3, 3, 1, 2

Here is a distribution:

x	0	1	2	3
P(X = x)	0.2	0.4	0.1	0.3

Test H_0 : The sample comes from
this distribution.

vs H_1 : The sample does not come
from this distribution.

Here are some sampled values:

1, 1, 2, 0, 1, 1, 1, 1, 3, 3,

1, 2, 0, 3, 1, 1, 3, 3, 1, 2

Collect the observed counts:

$$O_0 = 2, \quad O_1 = 10, \quad O_2 = 3, \quad O_3 = 5$$

(total is $n=20$)

Here is the distribution for H_0 :

x	0	1	2	3
$P(X = x)$	0.2	0.4	0.1	0.3

When H_0 is true, the expected counts are:

$$E_0 = (20)(0.2) = 4$$

$$E_1 = (20)(0.4) = 8$$

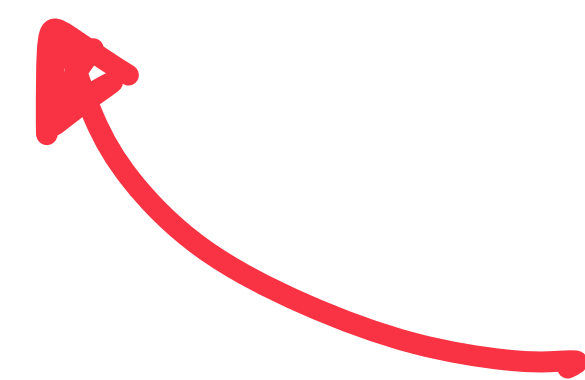
$$E_2 = (20)(0.1) = 2$$

$$E_3 = (20)(0.3) = 6$$

Consider the test statistic:

$$W := \sum_{i=0}^3 \frac{(O_i - E_i)^2}{E_i}$$

Claim: Under H_0 , W has roughly a $\chi^2(3)$ distribution!



number of
categories
minus 1

Claim: Under H_0 , W has roughly a $\chi^2(3)$ distribution!

This is a result of

- The Central Limit Theorem which says that

$$O_i = \sum_{j=1}^n I_{\{X_j=i\}}$$

gets normal in the limit.

Claim: Under H_0 , W has roughly a $\chi^2(3)$ distribution!

This is a result of

- The fact that a $N(0, 1)$ random variable squared has a $\chi^2(1)$ distribution.
- The fact that a $N(0, 1)$ random variable squared has a $\chi^2(1)$ distribution.

Claim: Under H_0 , W has roughly a $\chi^2(3)$ distribution!

This is a result of

- The fact that a sum of k independent $\chi^2(1)$ random variables has a $\chi^2(k)$ distribution

Claim: Under H_0 , W has roughly a $\chi^2(3)$ distribution!

However, it is complicated by the fact that $O_0 + O_1 + O_2 + O_3 = 20$, so these 4 random variables are not independent.

In general, for k categories and n observations,

$$W := \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \underset{\text{approx}}{\sim} \chi^2(k - 1)$$

for “large” sample sizes.

- “Large” n is not quite enough.
- If you have a large sample but one of the true probabilities is very small

x	0	1	2	3
$P(X = x)$	0.2	0.4	1×10^{-12}	the rest

then you still will have a difficult time getting observations of the outcome 2.

Rule of Thumb: Want the expected number (under H_0) in each category to be at least 5.

Back to the Original Example:

x	0	1	2	3
P(X = x)	0.2	0.4	0.1	0.3

When H_0 is true, the expected counts are:

$$E_0 = (20)(0.2) = 4$$

$$E_1 = (20)(0.4) = 8$$

$$E_2 = (20)(0.1) = 2$$

$$E_3 = (20)(0.3) = 6$$



Need more data!

Increased sample size to 100 and observed

$$O_0 = 18, \quad O_1 = 33, \quad O_2 = 12, \quad O_3 = 37$$

$$W := \sum_{i=0}^3 \frac{(O_i - E_i)^2}{E_i} \approx 3.458$$

Reject H_0 if W is “large”.

Increased sample size to 100 and observed

$$O_0 = 18, \quad O_1 = 33, \quad O_2 = 12, \quad O_3 = 37$$

$$W := \sum_{i=0}^3 \frac{(O_i - E_i)^2}{E_i} \approx 3.458$$

For a test of size $\alpha = 0.05$, “large” means $W > \chi_{0.05,3}^2 \approx 7.8147$.

Conclusion:

We fail to reject H_0 at level 0.05.

It appears that the data did come from the distribution

x	0	1	2	3
$P(X = x)$	0.2	0.4	0.1	0.3


```
> mysample<-sample(c(0,1,2,3),100,  
  replace=T,prob=c(0.2,0.4,0.1,0.3))  
> table(mysample)  
mysample  
  0    1    2    3  
18  33  12  37  
> obs<-c(18,33,12,37)  
> exp<-100*c(0.2,0.4,0.1,0.3)  
> W<-sum((obs-exp)^2)/exp)  
> W  
[1] 3.458333  
> qchisq(0.95,3)  
[1] 7.814728  
> |
```


H_0 : The sample comes from a binomial distribution with parameters 8 and 0.2.

You have n observations of
0's, 1's, ..., 8's.

Count up observations:

$$O_0, O_1, \dots, O_8$$

Expected numbers:

$$E_i = np_i \text{ where}$$

$$p_i = P(X = i) = \binom{8}{i} 0.2^i (1 - 0.2)^{n-i}$$

H_0 : The sample comes from the $N(0,1)$ distribution.

- Continuous data!
- Group data values into bins and do the test on this finite number of categories.
- Test can be sensitive to your choice of bins!
- Try a few different bin widths. Be leery of results if they are highly variable.